

Alice Gao

Researcher

Human-computer interaction • UI Understanding • Computational Interfaces • Human-AI interaction • Data-driven design tools

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Research Experience

Research Assistant / University of Washington

Jan 2024 – **Present** • Seattle, WA

- Developed novel machine-learning algorithm to train computer use agents (CUAs) to automatically perform automated usability testing, increasing CUA usability assessment accuracy by up to 45%.
- Synthesized automatic UI design metrics across literature to develop a new method of understanding UI design.
- Developed automatic pipeline to automatically compute design metrics for 10,000 collected websites across the globe and integrated into a research prototype.
- Led 10 user studies with UI/UX designers using a research prototype demonstrating new method of understanding UI design provides confidence and enables designers to provide more objective descriptions of design.
- Designed new interventions to reduce AI overreliance during AI-supported writing by up to 30% with 200+ participants in an online experimental study.

GRAIL / University of Washington

Sep 2021 – Jan 2024 • Seattle, WA

- Developed machine learning models to redirect attention and eye gaze for a more attentive video conferencing experience

Industry Experience

Capital One / Software Engineering Intern

Jun – Aug 2020 • New York City, NY

Created new endpoint and unit tests, laid framework for streaming data platform, and launched 2 new security groups.

Tezign / AI Design Lab Intern

Jun – Aug 2019 • Shanghai, China

Assisted in optimizing image retrieval to match similar designs and contributed to training a model in judging strong vs weak graphic designs for advertisements.

Education

University of Washington

Sep 2021 – *Expected Dec 2027*

MSc in Computer Science (2021 – 2023)

PhD in Computer Science (2023– **Present**)

Princeton University

2017 – 2021

B.S in Computer Science

Selected Publications

Alice Gao, Andrew N. Meltzoff, Maarten Sap, Katharina Reinecke. Framing an AI with Values Reduces AI Reliance in AI-supported Writing Tasks. ACM Conference on Fairness, Accountability, and Transparency (FAccT '26). Montreal, QC, Canada. June 2026.

doi.org/10.1145/3805689.3812369.

Alice Gao, Weixi Tong, Rishab Vempati, Katharina Reinecke, R. Benjamin Shapiro, Tianyi Zhang, Jason Wu. Training Computer Use Agents to Assess the Usability of Graphical User Interfaces. April 2026. arxiv.org/pdf/2604.26020.

Skills & Languages

Skills:

Machine learning • Post-training • Interviews (in-person and remote) • Survey Design • Usability testing • Thematic analysis • Statistical modeling • Data Analysis • Experimental design

Languages:

Python • R • Javascript • SQL • TypeScript

Frameworks / Tools:

PyTorch • scikit-learn • Pandas • React • Sveltekit • Flask • FastAPI • Figma